Grassland Management Workflow - Detailed Version

Contents

1. Download data with Order API

Use the Script

- Downloading single sites: *00_automated_download.R*
- Downloading multiple sites in a row: */00_planet_multiPoly_process.R*

- The API key can be found under the account page
- Adjust the AOI and date to be ordered
- Order in batches of up to 500 scenes per batch
- Some orders won't go through if a specific product is not available
  - Redo by removing scenes that are not available from order_url (see last part of script)
  - extract the failed urls and subset the asset batches variable to exclude the scence that is not available and run it again.
- download orders after successful order
- download after e.g. closing the session, retrieve all orders (end of script) and subset accordingly and download the images
  - see last part of script → get all order urls and subset accordingly
- Planet harmonization to Sentinel bands using harmonization tool
- Planet normalization → normalizing and harmonizing all sensors using harmonization tool
- Products from April 2022 onwards are 8 band or 4 band images
  - Everything before only 4 bands
  - each image has a UDM product. April 2022 is the only month that has both 4 and 8 band products, so that the 8 band images have been subset into 4 bands for simplification.

Downloaded data

- cloud cover of less than 70%
- period 2017–2025 for the months March to November
- stored in subfolders under CEPH_BASEDATA/SATELLITE/PLANET

After downloading images, ask Bart to move data to *CEPH_BASEDATA/SATELLITE/Planet*

2. Copy Data and Crop to AOI

2.1. PlanetScope

Scripts:

- 01_clip_raster_to_sites
- 02_check_planet_columns_rows.sh, 02_check_sr_udm.sh, 02_check_bands_valid.sh, 03_move_winterMonth.sh

After downloading the images and storing them in CEPH_BASEDATA use scripts to copy and crop all images to the respective AOI. Filter images also regarding ground control points, excessed through the metadata. Store images in the **folder *level2_sites_raw/site/standard and level2_sites_raw/site/test***. Check if all images have the same height & width and check if the number of udms equals the number of BOA images.

2.2. Sentinel-2

Sentinel images are in the following folders:

"/mnt/CEPH_PROJECTS/SAO/SENTINEL-2/SentinelVegetationProducts/FORCE/level2/X0003_Y0004"

"/mnt/CEPH_PROJECTS/SAO/SENTINEL-2/SentinelVegetationProducts/FORCE/level2/X0000_Y0004"

"/mnt/CEPH_PROJECTS/SAO/SENTINEL-2/SentinelVegetationProducts/FORCE/level2/X0000_Y0003"

"/mnt/CEPH_PROJECTS/SAO/SENTINEL-2/SentinelVegetationProducts/FORCE/level2/X0002_Y0005"

"/mnt/CEPH_PROJECTS/SAO/SENTINEL-2/SentinelVegetationProducts/FORCE/level2/X0004_Y0004"

"/mnt/CEPH_PROJECTS/SAO/SENTINEL-2/SentinelVegetationProducts/FORCE/level2/X0002_Y0003"

They are still in 10m resolution and tiled and have to be resampled:

Scripts: *10_s2_cubing.sh, 10_s2_cubing_qai.sh*

The script uses FORCE cubing algorithm to resample Sentinel-2 images to 3m resolution and cropping the tiles to a smaller area of interest.

Script: 11_check_s2_readability.sh

- *some images might be not accessible and not cubed because of different owner rights*

3. Geometric correction

Not all images align perfectly. In the terminal activate conda environment with *conda activate arosics* and run *python 04_arosics_exe.py*

Co-registered images are located in *level2_sites_raw/site/coregistered.* The code is run on every image, where there was no co-registration possible because e.g. the area was too small, images are either way copied and images are saved as .bsq, with a corresponding csv table with information about the coregistration and an error file that writes out all images that couldn't be co-registered and the corresponding error message. Images with NAS are not processed.

4. Cloud / shadow / snow / haze masking

4.1. Modified UDM2 mask

Planet provides a udm for every image, which contains information about cloud, shadow, haze and snow and a layer called "clear", which is everything that is not covered by anything mentioned before.

UDM masks are of low quality. Therefore 3 rules have been added to improve the masks:

1. For all clear layers == 1 check if SR files are not NA
2. Apply sieve filter to get rid of pixel clumps smaller than 1000 pixels

➔ ***Use Script:*** *05_LinuxOptimized_udm_improvement_script.R*

Modified udm mask are then saved in the level2_sites_raw/site/standard folder as *_udm2_mask.tif

Things to take care of:

- Rscript: Always add data type and NAflag for saving images and data type

Run *08_daily_images_V0.py*

to create daily composites with improved udm2 mask and blue band threshold. Output is used in the next step.

4.2. Improved shadow mask and cloud buffering

To get also the fade-outs of the clouds, buffer the clouds. To improve the shadow mask, calculate a color difference measure of the RGB bands using the script */06_colorMeasure.R*

Planets shadow mask often misses shadows, so I created a new mask using cloud-free PlanetScope mosaics from July for each year and a difference measure of the RGB bands based on the previous step (daily images used for creating cloud free mosaics). Cloud shadows reduce the reflectance and appear darker in the RGB channels. If this measure doesn't exceed a certain threshold, it should be considered as shadow. In combination with comparing the NIR band of each image with the NIR band of the reference mosaic scaled by a dark factor that was derived from looking at the ratio between nir_target / nir_reference and a NIR band threshold, where cloud shadows appear very dark.

The formular

Color difference measure: $$|\mathbf{R} - \boldsymbol{G}| + |\mathbf{R} - \boldsymbol{B}| + |\mathbf{G} - \boldsymbol{B}|$$

Shadow mask:

$$\left(\mathrm{NIR}_{\mathrm{target}} < (\mathrm{ref}_{\mathrm{summer}} * \mathrm{dark\ factor})\right) \,\&\, \left(\mathrm{NIR}_{\mathrm{target}} < \mathrm{NIR}_{\mathrm{thres}h}\right) \,\&\, (\mathrm{RGB}_{\mathrm{diff}} < \mathrm{RGB}_{\mathrm{thresh}})$$

- ➔ *Run: 07_shadow_cloud_buffer.py*

Shadow masks are then newly created using the abovementioned algorithm and combined with the old shadow mask. The results are saved in level2_sites_raw/site/standard as *_udm2_buffer.tif

5. PLANET DAILY IMAGES

Since several observations are taken on the same date, Planet data has been composited using the mean value of all available valid observations. For that, the following structure was created:

FORCE/level2_sites_daily/..

..00/ masked and composite Planet data using the original udm2 masks

..01/ masked and composite Planet data using the adapted udm2 masks (*udm2_mask.tif)

..03/ masked and composite Planet data using the adapted udm2 masks with the new shadow mask (*_udm2_buffer.tif)

Run *08_daily_images_V3.py* to create daily images with new shadow and cloud buffer mask.

08_daily_images_V0.py and *08_daily_images_V1.py* to create daily images with different masks.

Planet has it's own composite tool in the ORDER API but

- Problem is that there are still several products even after using the strip_id as compsite option
- *Group_by: strip_id* create a composite is created using the last scene in the order
- Composite option *group_by: area* creates one composite for all images in the order

6. NDVI

PLANET

- Planet images were masked with the modified UDM mask, then the NDVI was calculated for each image
- Stored under level3_sites/indices for the different versions in level2_sites_daily
- Use the level2_daily images as input to calculate NDVI
- Then create ndvi_tss.bsq with header and dates.txt files for each year
    - Check header again for empty spaces and correct it

Planet scripts: 09_calculate_ndvi.sh, 09_ndvi_tss.sh

SENTINEL

The Sentinel NDVI TSS was using FORCE for the tiles

X0000_Y0003, X0002_Y0005, X0003_Y0004, X0004_Y0004, X0002_Y0003

1. unstacked using the *12_unstack_ndvi.R*
    a. stored Sentinel NDVI 10m images under *FORCE/SEN2_NDVI*
2. Cube NDVI to 3m using *13_s2_cubing_ndvi.sh*
    a. To resample to 3m and crop to aoi
3. Clip with *13_clip_s2_ndvi_to_extent.R,*
    a. Clip to mask extents of the sites
4. Mosaic NDVI with *14_mosaic_s2_sites_ndvi.sh,*
    a. Mosaic sites because for some sites tile number > 1
5. Crete ndvi tss for Sentinel only with *15_ndvi_tss_sen2.sh*
6. Create ndvi tss for Sentinel and Planet using *15_ndvi_tss_pla_sen2.sh*

➔ Results stored under level3_sites/indices/site/

7. Additional data

7.1. GDDs, Temperature and Snow

- Run *scripts/gdd/01_process_temperature.R* to create yearly temperature raster for min, max and mean over South Tyrol
- Calculated first DOY when 5°C in 5 consecutive days for each raster cell from temperature average data with *02_firstover5.sh*
- Calcualte first frost of the year with different mowing windows, 2,3 or 5 consecutive days of temperature < 0 with *02_firstFrost.sh*
- Calculate GDD for South Tyrol with *03_GDD_cum_ST.py* using the *first_over_5.tifs* as input for the starting date of GDD
    - Script calculates daily temperature
    - Cumulative daily temperature
    - Cumulative GDDs
- Calculate first DOY with GDD over a certain value using *gdd/04_firstDOY_GDD.sh*

Additional scripts:

- Compare DOY of condition to NDVI, with first DOY when NDVI > 0 for Mals Heath and created agreement map for all years. *firstposNDVI.sh*
- Check snow layers from VIIRS & MODIS 250m and compared to temperature condition 5°. Calculated first DOY when snow fraction < 100% in 3 consecutive days.

7.2. LAFIS & WBS

7.2.1. LAFIS

Run *lafis_wbs/scriptToClipLafis.R*

Lafis shapfiles for 2017:2025 each contain different column names.

1. First harmonise lafis files by extracting the CUAA's + field ids from each file and create a new column FID_std + create a new unique id: CUAA + FID_std

2017, 2018, 2023:

- FID_std = FID, DESCR = DESCRIPT_1

2019:

- FID_std = PROG_POLIG, DESCR = DESCRIPT00

2020, 2021, 2022, 2024

- FID_std = "PROG_POLIG", DESCR = "DESCR_DE"

2025

- FID_std = FID_1, DESCR = DESCRIPT_1

2. In the code validate the polygons for incorrect geometries because some of the fields have double vertices and incorrect geometries.
3. Clip Lafis to Mals Heath and to the respective zone
   - Clip to zones → creates multiple polygons with same id but in different zones because some polygons overlap the two zones
   - calculate area and calculate how much of the area is located in which zone. If it's less than 1/7 of the total area, join polygon back to original zone, if bigger create new polygon with different zone but same unique id

4. Transform into EPSG 32632
5. Save all lafis files as:
   gis/ST/lafis/{year}/lafis_grassland_{year}_{version}_32632_zones.shp

7.2.2. WBS

Data

- 2021, 2022, 2023 PDF with parcel numbers and for 2023 underlined applied fields
- Excel sheet with parcel numbers, location, money, area and zone for 2021 & 2022
- LAFIS 2021, 2022, 2023
- ParcelsAggregate_polygon.shp

Workflow

Special case 2023

- For 2023 there was only the PDF available with the underlined fields that have applied to the program
- In the PDF the parcel numbers are displayed
- Took the parcelsAggregate_polygons.shp and searched for each underlined parcel number in the corresponding zone
- Joined them with the lafis shapfiles by creating a centroid in each field and intersected it with the corresponding lafis field

2021, 2022

--> PDFs + csv tables available with parcel number and name of applier

- Take the ParcelsAggregate_polygon.shp and create a new Unique id consisting of region, parcel number and zone
- Did the same for the csv tables
- Intersected the WBS 2021 and 2022 with the ParcelsAggregate_polygon.shp by the unique identifier
- Checked the difference between the original csv table's parcel numbers and those in the parcel shapefile for each year.
    - Filtered out the column '*PART_CODIC*' in the parcel polygon shapefile by comparing the unique id to the csv table file
- Checked the missing parcel numbers in the Excel spreadsheet again and tried to find them in the PDFs. Visually checked again with the parcel shapefile in QGIS.
- Corrected some missing numbers due to bad formatting and added comments to the Excel spreadsheet to ask M. Joos.
- Call with M. Joos to understand why some fields are missing in LAFIS and to check unclear fields.
- Another check for missing data after the call.

- Left out the fields listed in the application under 'Effektive Nutzung', as they are not in the LAFIS shapefile and are very small.
    - Several polygons overlap, so it is not possible to make a clear assignment.
- Some fields are not found in the 2021 or 2022 LAFIS shapefiles but are present in the 2025 shapefile.
    - Created two new columns in the LAFIS 2021, 2022 and 2025 shapefiles: 'wsb' and 'Voll' to identify missing parcels and evaluate whether the LAFIS polygon is fully covered by the full parcel shapefile polygon.
- Merged everything back together and created two new shapefiles:
    - 'Wiesenbrueter_LAFIS_2021.shp' and 'Wiesenbrueter_LAFIS_2022.shp'.
- Adding missing CUAAs and corrected some according to the Excel Sheet applicants

For all wbs shapefiles

- Run *lafis_wbs/wbs_pdf_to_points.R*
    - Tries to identify potential polygons in parcelpolygon shapfile with the help of the csv tables
    - Clippe all wbs shapefiles to zones shapefile
- *Run lafis_wbs/join_wsb_lafis.R to* create a wbs data base with lafis polygons
    - Create a centroid for each polygon and intersected the point with the corresponding lafis shapfiles: *gis/ST/lafis/{year}/lafis_grassland_{year}_{version}_32632_zones.shp*
- Data for 2020 from Joachim Winkler, tried to identify them in the parcels shapefile and identified the corresponding lafis polygon à not all of them could be clearly identified and are missing in LAFIS
- Select only AP2, AP4, AP5, AS

Problems for statistical analysis

- LAFIS and parcel shapefile outlines do not match
    - In some cases, the LAFIS field is bigger than the actual field size as shown in the PDF or in the province parcel shapefile
    - Not displaying the correct area size
- CUAA of LAFIS and excel sheet are not always the same due to multiple field managers, owners and program applicants

Tried to align the CUAA with the ones in the excel sheet but for 2024 and 2025 we don’t have that information, so there could be some mismatches.

Not sure, if the field ids are always the same over the years.

7.3. Site masks

Generate a raster layer for each site using *000_create_extent_mask.R* that uses an exemplary image from level2_sites_raw as input image to get the width and height of the raster to pass it to the masks.

Cube all additional layers to resample to 3m, and clip to same raster width and row as the boa data:

1. 16_cube_DOY.sh
2. 16_cube_frost.sh
3. 16_cube_lafis.sh
4. 16_cube_masks.sh

8. Statistical analysis

- Create statistics for LAFIS v WBS
- Create statistics for PlanetScope and Sentinel data availability for each test site
- Use the scripts:

- ***analysis/analysisWBSLAFIS_presentation.R***
- ***analysis/S2_planet_availability_new.R***
- ***analysis/sentinel_planet_radiometric_comparison.R***

OLD

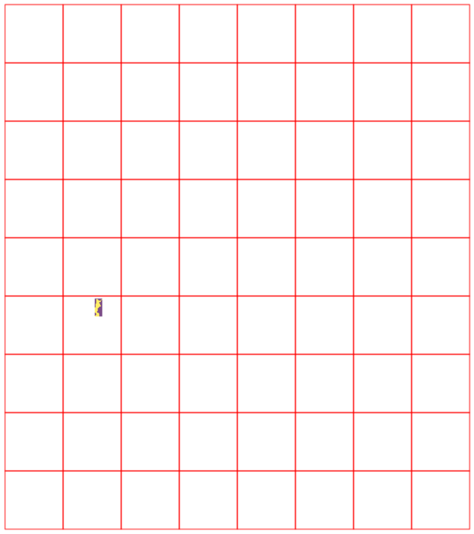
FORCE Cubing

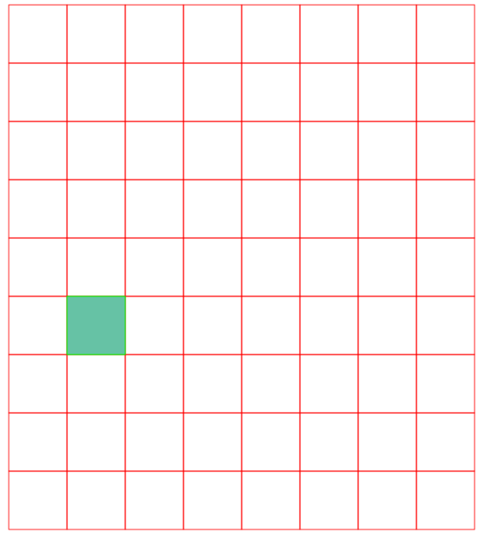
The data is processed into the FORCE data cube format.

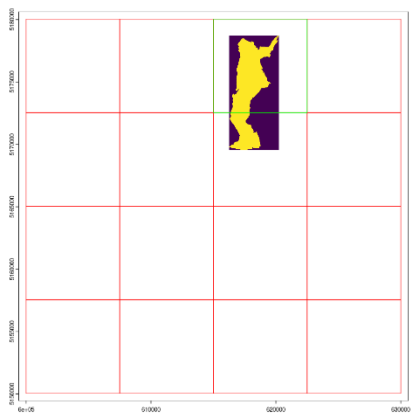
1. Create the datacube file with Environtwin/Scripts/create_data_cube_3m.R

DATACUBE GENERATION

The data cube was generated based on the Sentinel 2 tiling grid from FORCE. The tile X0000_Y0004 was selected, where the Mals Heath is located. The tile was then divided into 4x4 grid, then selecting the cell where Mals Heath is located.







The coordinates of the upper left corner of the cell were then extracted and used as the input for the FORCE cube. To generate the cube, a 4000 x 4000m quadrat was fitted starting at the upper left corner to get only a one tiled cube X-001_Y-001 with the 3x3m resolution PlanetScope data in it.

CUBE ORGANIZATION

Level2_raw = Planet + Sentinel at 3m – original images

Level2_daily

Scripts

- 11_dailyPlanetMask_0*.py

- 01 – Planet data mosaiced to one image per date using mean for reflectance values and sum for modified udm masks and then turned into binary mask with 0 and 1

- 02 – Planet data mosaiced to one image as above and modified udm mask + blue band threshold. Blue band > 900 then cloud / haze / snow

Level3

Scripts

  - 07_s2_cubing_ndvi.sh and 08_*.sh
  - Use level2_daily images to calculate NDVI

- *Indices*
  - *00:* original udm mask NDVI images and time series
  - *01:* modified mask NDVI images + time series
  - *02:* modified mask + blue band threshold NDVI images + time series
  - *11:* mean and median NDVI based on 01
- *mosaics*

  PLANET_MOSAIC_4BANDS_PERIOD_MH

  PLANET-SENTINEL_MOSAIC_4BANDS_PERIOD_MH

Level4/mowing

➔ mowing detection results for the related level3/indices folders
- *00*
- *01*
- *02*
- *11*
- *12*
- *22*

CUBING AND FURTHER PROCESSING

- Run *01_* scripts to cube the different products.
  - -Must be run separately. Otherwise, it will stack everything together.
- Run *02_* to rename the products to _BOA.tif and _udm2.tif and subset the 8 band product to 4 bands
- Run *03_* scripts to check for differences in size, readability and number of SR files and UDM files is the same.
- Move and delete files that are incomplete or in winter into another folder with *04_delete_incomplete.sh and 05_move_winterMonths*
  - Incomplete files are files with missing udm or boa file.

- Rename bands for udm2 and boa products with 06_ scripts
- Cube NDVI S2 from S2_NDVI to 3m planet cube
- Calculate NDVI for Planet images
- Generated NDVI mosaics for Seiser Alm and Armentarawiesen for Ruths median and mean calculation → field level mowing detection

Several other preprocessing steps have been added, e.g. the calculation of GDD to improve the mowing detection and the first frost layers. All calculated results were also cubed and are located under ***FORCE/masks.*** Further information on the calculations of these layers in the later sections.

Other scripts:

- *get_image_ids* to get the first part of the Planet data
    - Can be used to check for missing data and to download them again via the automated_download.R script
- *planet_cubing_missing*

Further considerations:

- Correct merging of level2_raw products?
- Use Planets daily products using compositing tool
- Udm2 mask improvement with blue band threshold
- Geometric alignment of all images given?
- How is geometric alignment with Sentinel images?